Course 2: LoRa Basics Modem

LBM Architecture & Integration

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USP Zephyr Training Series

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Course Overview

What You'll Learn

- LoRa Basics Modem Architecture
  - Event-driven programming model
  - Modem API and services
  - Stack lifecycle management
- Advanced Features
  - Low power optimization
  - GNSS geolocation
  - LoRaWAN Relay
- Hands-On Labs (5 labs)
  - Lab 2.1: Build and run periodical uplink
  - Lab 2.2: Custom event handler
  - Lab 2.3: Low power configuration
  - Lab 2.4: GNSS geolocation
  - Lab 2.5: Relay TX configuration

What is LoRa Basics Modem?

Semtech's Official LoRaWAN Stack

Definition

LoRa Basics Modem (LBM) is Semtech's production-ready LoRaWAN stack implementing the complete MAC layer protocol.

Features:

- LoRaWAN 1.0.4 certified
- Event-driven architecture
- Low power optimized
- Built-in services:
  - GNSS scanning
  - LoRaWAN Relay
  - FUOTA support
  - Class A/B/C

Version Info:

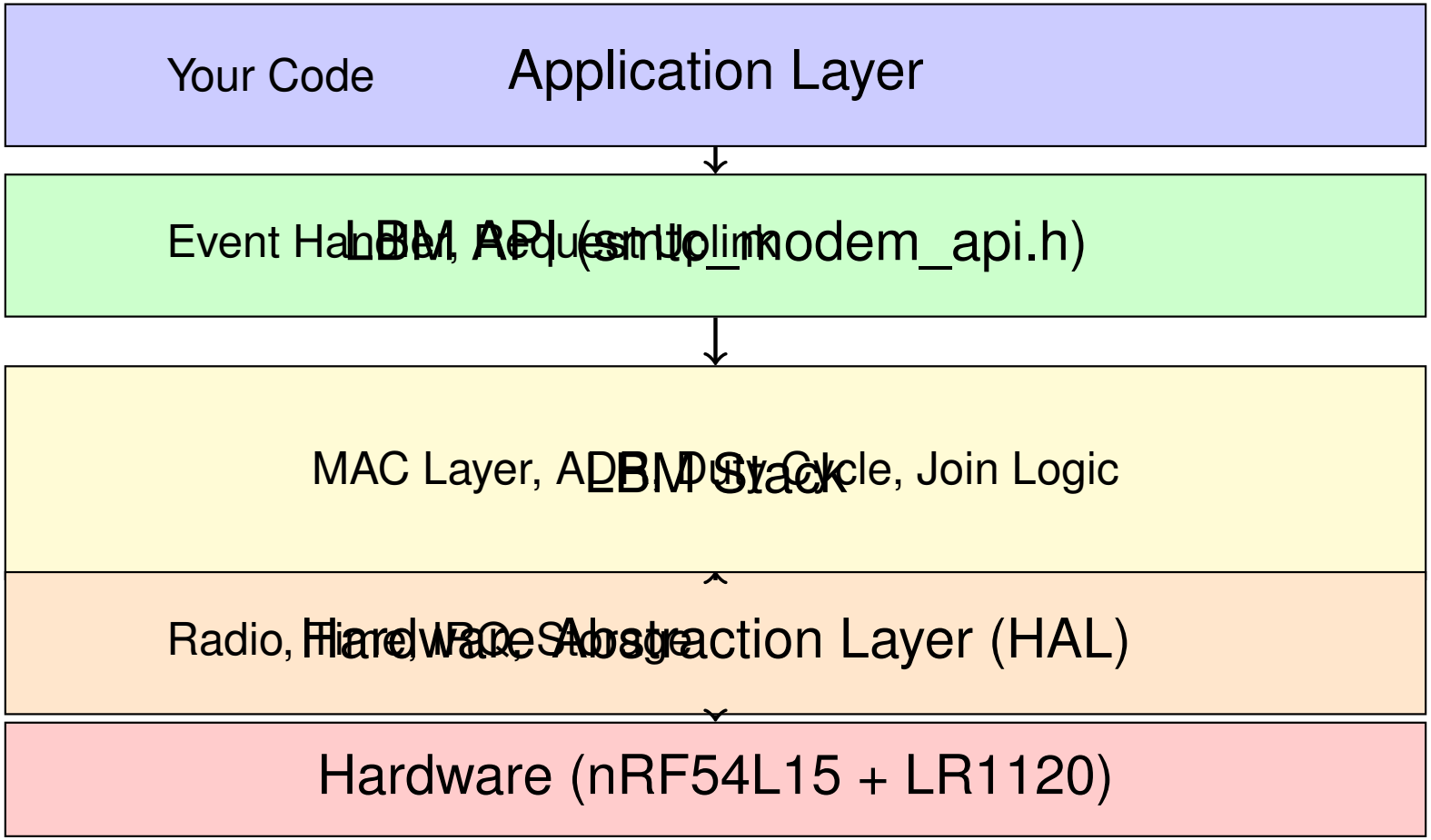
- Current: v4.9.0
- License: BSD-3-Clause
- Maintained by Semtech
- Used in production

USP Integration:

- Zephyr RTOS port
- Device tree config
- West workspace

LBM Architecture

Software Stack Layers



Key Point: Application code only interacts with LBM API, not hardware directly.

Event-Driven Programming

How LBM Communicates with Application

Event Loop Pattern

LBM uses events to notify the application of state changes and incoming data.

```c
void event_handler(void)
{
    smtc_modem_event_t event;
    uint8_t pending_count;

    while (smtc_modem_get_event(&event, &pending_count) == SMTC_MODEM_RC_OK) {
        switch (event.event_type) {
        case SMTC_MODEM_EVENT_JOINED:
            LOG_INF("Network joined!");
            break;

        case SMTC_MODEM_EVENT_TXDONE:
            LOG_INF("TX done: %u", event.event_data.txdone.status);
            break;

        case SMTC_MODEM_EVENT_DOWNDATA:
            LOG_INF("Downlink received");
            handle_downlink(&event);
            break;
        }
    }
}
```

LBM Event Types

Complete Event Reference

Event	Description
JOINED	Network join successful
JOINFAIL	Join attempt failed
TXDONE	Uplink transmission complete
DOWNDATA	Downlink data received
LINK_CHECK	Link check result available
CLASS_B_STATUS	Class B beacon status
ALARM	Alarm timer expired
GNSS_SCAN_DONE	GNSS scan completed
RELAY_TX_SYNC	Relay TX synchronized
FUOTA	Firmware update event

Best Practice: Process events in a dedicated thread to avoid blocking modem engine.

LBM API Basics

Common Operations

Initialization:

```
smtc_modem_init(event_handler);
smtc_modem_set_region(STACK_ID, SMTC_MODEM_REGION_EU_868);
```

Join Network:

```
smtc_modem_join_network(STACK_ID);
```

Send Uplink:

```
uint8_t payload[] = {0x01, 0x02, 0x03};
smtc_modem_request_uplink(STACK_ID,
                          2,       // Port
                          false,   // Unconfirmed
                          payload,
                          3);      // Length
```

Get Status:

```
smtc_modem_status_mask_t status;
smtc_modem_get_status(STACK_ID, &status);
if (status & SMTC_MODEM_STATUS_JOINED) {
    LOG_INF("Connected to network");
}
```

Power Consumption Breakdown

Where Does Power Go?

State	Current	Duration	Energy
Sleep	25 µA	95%	0.57 mAh/day
TX (SF7, +14dBm)	100 mA	41 ms × 24	0.027 mAh/day
RX1 Window	15 mA	100 ms × 24	0.010 mAh/day
RX2 Window	15 mA	100 ms × 24	0.010 mAh/day
		Total (60s interval):	**0.62 mAh/day**

Battery Life Calculation (2000 mAh):

$$\text{Life} = \frac{2000 \text{ mAh} \times 0.7}{0.62 \text{ mAh/day}} = 2258 \text{ days} \approx 6.2 \text{ years}$$

Key Insight

Sleep current dominates for long-lived sensors! Optimize sleep mode first.

Low Power Strategies

Extend Battery Life

1. **Increase Uplink Interval**
   - 60s $\rightarrow$ 300s (5min): 5× battery life improvement
   - Consider application requirements
2. **Optimize Spreading Factor**
   - SF12: 991 ms ToA, but better sensitivity
   - SF7: 41 ms ToA, but requires good link
   - Use ADR to find optimal balance
3. **Reduce Payload Size**
   - Fewer bytes = shorter ToA
   - Use efficient encoding (binary, not JSON!)
4. **Disable Unnecessary Features**
   - No LED blinking
   - Disable UART console in production
   - Use confirmed uplinks sparingly
5. **Hardware Optimization**
   - Enable Zephyr power management

GNSS with LR1120/LR1121

Satellite-Based Positioning

How It Works:

1. LR1120 scans for GPS/BeiDou satellites
2. Generates NAV (navigation) message
3. Send NAV to LoRa Cloud via LoRaWAN
4. Cloud solver returns position

Advantages:

- No dedicated GPS receiver needed
- Passive scanning (no TX)
- Works with obstructed view

Accuracy:

- Outdoor, clear sky: 10-30m
- Near window: 30-100m
- Indoor: May fail or >100m

Scan Modes:

STATIC: Device not moving, longer scan

MOBILE: Device moving, shorter scan

Time to Fix:

- With assistance: 30-60s
- Cold start: 2-5 min

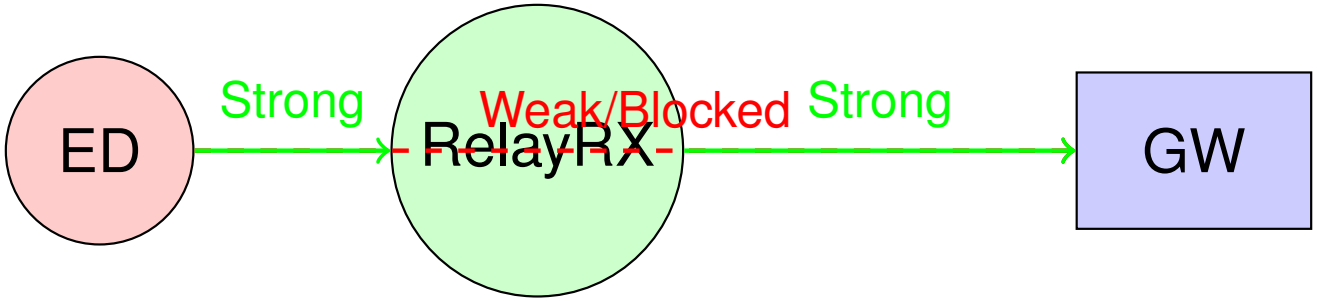
GNSS Implementation

Code Example

```
// Configure GNSS
smtc_modem_gnss_set_constellation(STACK_ID,
    SMTC_MODEM_GNSS_CONSTELLATION_GPS_BEIDOU);

// Set assistance position (approximate location)
smtc_modem_gnss_assistance_position_t assist = {
    .latitude = 37.7749,
    .longitude = -122.4194,
};
smtc_modem_gnss_set_assistance_position(STACK_ID, &assist);

// Start scan
smtc_modem_gnss_scan(STACK_ID, SMTC_MODEM_GNSS_MODE_STATIC);

// In event handler:
case SMTC_MODEM_EVENT_GNSS_SCAN_DONE:
    uint8_t nav_message[255];
    uint16_t nav_size;

    smtc_modem_gnss_get_event_data_scan_done(STACK_ID,
                                             nav_message,
                                             &nav_size);

    // Send NAV to cloud solver
    smtc_modem_request_uplink(STACK_ID, 192, false,
                              nav_message, nav_size);
    break;
```

LoRaWAN Relay

Extend Network Coverage



Use Cases:

- Indoor sensors with outdoor gateway
- Extending range in difficult terrain
- Connecting devices in basements/underground

Types:

Relay RX: Mains-powered, Class C, forwards uplinks

Relay TX: Battery-powered end device, relayed by RX

Relay TX Configuration

Enable Relay for End Device

```
// After join, enable Relay TX
smtc_modem_relay_tx_enable(STACK_ID,
    SMTC_MODEM_RELAY_TX_ACTIVATION_MODE_ENABLE);

// In event handler:
case SMTC_MODEM_EVENT_RELAY_TX_SYNC:
    LOG_INF("Synchronized with Relay RX");
    LOG_INF("  Relay DevAddr: 0x%08X",
            event.event_data.relay_tx_sync.relay_dev_addr);
    break;

case SMTC_MODEM_EVENT_TXDONE:
    // Check if uplink was relayed
    smtc_modem_relay_tx_uplink_info_t info;
    smtc_modem_relay_tx_get_last_uplink_info(STACK_ID, &info);

    if (info.was_relayed) {
        LOG_INF("Uplink was RELAYED");
        LOG_INF("  Via: 0x%08X", info.relay_dev_addr);
    } else {
        LOG_INF("Direct uplink (not relayed)");
    }
    break;
```

Power Benefit: Relay TX can use lower TX power, saving battery!

Lab 2.1: Build and Run Periodical Uplink

Your First LBM Application

Objective: Build, flash, and run the LBM periodical uplink sample.

Steps:

1. Configure credentials in device tree overlay
2. Build: `west build -p -b xiao_nrf54l15_nrf54l15_cpuapp \ -shield semtech_lr1120mb1dis samples/usp/lbm/periodical_uplink`
3. Flash: `west flash`
4. Monitor: `minicom -D /dev/ttyACM0 -b 115200`
5. Verify on network server (TTN/ChirpStack)

Expected Output:

- Join successful within 10 seconds
- Uplinks every 60 seconds
- RSSI/SNR values logged

Deliverables:

- Serial log showing join + 10 uplinks
- Network server screenshot

Lab 2.2: Custom Event Handler

Advanced Event Processing

Objective: Implement custom logic to track success rate and detect failures.

Features to Implement:

1. Track uplink success rate (confirmed vs total)
2. Alert if 3 consecutive failures
3. Log RSSI/SNR of downlinks
4. Automatic ADR profile switching based on link quality

Example Alert Logic:

```
static uint8_t consecutive_failures = 0;

case SMTC_MODEM_EVENT_TXDONE:
    if (event.event_data.txdone.status == SMTC_MODEM_EVENT_TXDONE_CONFIRMED) {
        consecutive_failures = 0;
    } else {
        consecutive_failures++;
        if (consecutive_failures >= 3) {
            send_alert_uplink();  // Notify server
        }
    }
    break;
```

Deliverables:

Lab 2.3: Low Power Configuration

Optimize for Battery Life

Objective: Configure LBM for minimum power consumption.

Tasks:

1. Build with low power config: `prj_lowpower.conf`
2. Measure power with Nordic Power Profiler Kit II
3. Compare different uplink intervals (60s, 300s, 900s)
4. Calculate battery life for each configuration

Measurements to Take:

- Sleep current (should be < 30 µA)
- TX current and duration
- RX window current and duration
- Average current per uplink cycle

Expected Results:

Interval	Avg Current	Battery Life
60 seconds	500 µA	166 days
300 seconds	100 µA	833 days
900 seconds	45 µA	1,851 days

Lab 2.4: GNSS Geolocation

Satellite-Based Positioning

Objective: Implement GNSS scanning and solve position with LoRa Cloud.

Prerequisites:

- LR1120 or LR1121 radio (has GNSS capability)
- Clear view of sky (outdoors or near window)
- LoRa Cloud account with MGMS token

Steps:

1. Configure GNSS constellation (GPS + BeiDou)
2. Set assistance position (approximate location)
3. Start GNSS scan after join
4. Send NAV message to LoRa Cloud
5. Receive solved position as downlink
6. Visualize on map (Python script provided)

Deliverables:

- Serial log showing NAV message and solved position
- HTML map with position marker

Lab 2.5: Relay TX Configuration

Extended Range with Relay

Objective: Configure device as Relay TX (relayed end-device).

Prerequisites:

- Two devices: one Relay TX (this lab), one Relay RX
- Network server supporting LoRaWAN Relay

Setup:

1. Configure and deploy Relay RX near gateway
2. Configure this device as Relay TX
3. Enable Relay TX mode after join
4. Verify uplinks are relayed (check metadata)
5. Measure power savings from reduced TX power

Verification:

- Serial log shows "Relay TX synchronized"
- Uplink events indicate "was relayed"
- Network server shows relay metadata

Deliverables:

- Relay TX code

Course 2 Summary

Key Takeaways

You've Learned:

- LBM Architecture: Event-driven, modular, production-ready
- Event Processing: How to handle JOIN, TXDONE, DOWNDATA events
- Low Power: Optimize sleep, interval, SF, payload for max battery life
- GNSS: Satellite positioning without dedicated GPS receiver
- Relay: Extend coverage using relay RX/TX devices
- Practical Skills: Build, optimize, and deploy LBM applications

Best Practices:

- Always use event-driven pattern (don't poll!)
- Test power consumption early in development
- Use ADR for optimal SF selection
- Consider Relay for indoor/difficult deployments

Next Course: Course 3 - USP RAC Architecture (Multiprotocol)

Thank You!

Questions?